ECD 416

Project Report On

MEDICAL IMAGING AI FOR BRAIN TUMOR DIAGNOSIS

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Under the Supervision of

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Certificate

This is to certify that this project report entitled **"Medical Imaging AI for Brain Tumor Diagnosis "** *is a bonafide record of the work done by* ***Adarsh A S*** (KTE22EC005), ***Anjali Tomson Joseph***(KTE22EC019), ***Anjana T*** (KTE22EC021), and ***Anna Rose V Vinod***(KTE22EC022) *under our guidance towards the partial fulfillment of the requirements for the award of the degree of* ***Bachelor of Technology*** *in* ***Electronics and Communication Engineering*** *of A P J Abdul Kalam technological university, Trivandrum, during the year 2025-26 and that this work has not been submitted elsewhere for the award of any degree.*

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ABSTRACT

Brain tumor detection using Magnetic Resonance Imaging (MRI) is an important task in medical diagnosis, as early identification of tumors can significantly improve treatment outcomes. Manual analysis of MRI scans can be time-consuming and prone to errors, especially when handling large volumes of medical data. To address these challenges, this project proposes an automated brain tumor detection system using deep learning techniques.

In the proposed system, MRI images first undergo preprocessing steps such as resizing and normalization to ensure consistency and improve model performance. A convolutional neural network based on the ResNet50 architecture is then used for classification, where the input MRI image is categorized into tumor or non-tumor classes. The system is designed to detect common types of brain tumors such as glioma, meningioma, and pituitary tumors. The classification model provides prediction results along with a confidence score using the Softmax function.

Following classification, a segmentation model is applied to accurately identify and highlight the tumor region within the MRI image. This helps in visualizing the exact location and shape of the tumor, which can assist in further medical analysis and treatment planning.Both models are deployed on a Raspberry Pi connected to a touchscreen display, and a web-based interface enables users to upload MRI images and view the classification and segmentation results.

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Chapter 1

Introduction

Brain tumors are among the most serious neurological disorders affecting the human brain. They occur due to the abnormal growth of cells within brain tissues, which can interfere with normal brain functions such as memory, movement, and sensory processing. Early detection of brain tumors is extremely important for effective treatment planning and improving patient survival rates. If diagnosed at an early stage, appropriate medical treatment can significantly reduce complications and improve the quality of life for patients.

Medical imaging techniques play a vital role in identifying and diagnosing brain abnormalities. Among these techniques, Magnetic Resonance Imaging (MRI) is widely used because it provides highly detailed images of brain tissues without exposing patients to harmful radiation. MRI scans offer excellent soft tissue contrast, making them suitable for detecting abnormalities such as tumors, lesions, and other structural changes in the brain. These detailed images allow doctors to study brain structures carefully and identify suspicious regions that may indicate tumor growth.

Traditionally, MRI scans are examined manually by radiologists or medical experts. This process requires considerable expertise and time, especially when large numbers of scans need to be analyzed. In addition, brain tumors can vary greatly in size, shape, and location, which makes the detection process challenging. Human interpretation may sometimes lead to delays or inaccuracies, particularly when the workload is high. Therefore, there is a growing need for automated systems that can assist medical professionals in analyzing medical images more efficiently.

Recent advancements in Artificial Intelligence (AI) and deep learning have significantly improved the field of medical image analysis. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated excellent performance in detecting patterns and extracting meaningful features from complex images. These models are capable of automatically learning important features from MRI images and identifying abnormal regions that may correspond to tumors.

The main aim of this project is to develop an automated brain tumor detection system using deep learning techniques. The proposed system integrates image preprocessing, tumor segmentation, and classification to analyze MRI brain images and determine whether a tumor is present. The system is designed to detect common types of brain tumors such as glioma, meningioma, and pituitary tumors. The MRI image is first preprocessed through resizing and normalization to ensure compatibility with the deep learning models.

After preprocessing, the MRI image is first passed through a deep learning classification model based on the ResNet50 architecture to determine the class category to which the image belongs. The classification output is generated using the Softmax function, which provides probability scores for each class and identifies the most likely category. Following classification, a segmentation model is applied to identify and highlight the region of interest within the MRI image. Segmentation assists in accurately localizing the abnormal tissue region, enabling better visualization and analysis of the affected area.

For practical implementation, the system is deployed on a Raspberry Pi platform to create a compact and portable diagnostic tool. A web-based interface developed using HTML, CSS, and JavaScript allows users to upload MRI images and view the diagnostic results.

The Raspberry Pi is connected to a touchscreen display that enables interactive operation of the system. Users can upload MRI images, visualize segmentation results, and view classification outputs directly through the touchscreen interface. This system demonstrates how deep learning models can be effectively integrated with embedded hardware to provide an intelligent and portable solution for automated brain tumor detection.

Chapter 2

Literature Survey

The literature survey provides an overview of recent research developments in the field of automated brain tumor detection using Magnetic Resonance Imaging (MRI). In recent years, the integration of artificial intelligence (AI) and deep learning techniques has significantly improved the accuracy and efficiency of medical image analysis. Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated remarkable performance in identifying complex patterns in MRI images, enabling automated tumor detection and classification.

Several researchers have proposed different deep learning architectures and frameworks to enhance tumor detection, segmentation, and classification performance. These approaches aim to assist radiologists by reducing diagnostic time, minimizing human error, and improving overall reliability in medical diagnosis. This chapter reviews five significant research papers that contribute to advancements in brain tumor detection using MRI images. Among these studies, the work by *Abdusalomov et al. (2023)* [1] serves as the **base paper** for this project, as it provides a comprehensive overview of deep learning techniques for MRI-based tumor detection.

[1] Base Paper — Abdusalomov, A. B., et al. (2023) [1] presented a deep learning-based framework for automated brain tumor detection using MRI images. The authors explored various convolutional neural network architectures and demonstrated how deep learning models can effectively identify tumor regions from MRI scans. The study focused on improving diagnostic accuracy through advanced image processing and feature extraction techniques.

The proposed system utilized deep neural networks to automatically extract meaningful features from MRI images without relying on manual feature engineering. The researchers emphasized the importance of preprocessing steps such as normalization and image enhancement to improve the performance of deep learning models. The study highlighted that CNN-based models can significantly outperform traditional machine learning approaches in medical image analysis tasks.

Experimental results showed that the proposed deep learning approach achieved high accuracy in identifying brain tumors and detecting abnormal regions in MRI scans. The authors concluded that automated tumor detection systems can assist medical professionals by providing faster and more consistent diagnostic results. This research provides a strong theoretical foundation for developing AI-based diagnostic tools and serves as the base reference for the present project.

[2] K. Neamah et al. (2024) [3] proposed an improved deep learning framework for brain tumor classification using MRI images. The study introduced an enhanced version of the ResNet50 architecture designed to improve feature extraction capability and classification accuracy. Residual networks are known for their ability to mitigate the vanishing gradient problem in deep neural networks by introducing shortcut connections between layers.

The improved ResNet50 model was trained using MRI datasets to classify brain tumors effectively. The study demonstrated that the enhanced architecture achieved higher classification accuracy compared to conventional CNN models. The results indicated that residual learning significantly improves the performance of deep neural networks in complex medical image classification tasks. This work highlights the effectiveness of ResNet-based architectures in automated tumor detection systems.

[3] R. Singh et al. (2024) [6] proposed an ensemble deep learning framework for enhanced brain tumor classification using high-resolution MRI images. The proposed method combined multiple deep learning architectures, including ResNet50 and EfficientNet-B7, to improve classification accuracy.

The ensemble approach allowed the model to leverage the strengths of different neural network architectures, enabling better feature extraction and improved generalization capability. Experimental results demonstrated that the ensemble model achieved superior performance compared to individual deep learning models. The study concluded that combining multiple deep learning models can significantly enhance the reliability and robustness of automated medical image classification systems.

[4] P. Kumar Tiwary et al. (2025) [7] presented a deep learning-based approach for brain tumor segmentation using an EfficientNet-enhanced U-Net architecture. The study focused on accurately identifying tumor regions within MRI images through semantic segmentation techniques.

The proposed model utilized the encoder–decoder structure of the U-Net architecture to capture both local and global features of MRI images. EfficientNet was integrated into the architecture to enhance feature extraction capability and improve segmentation accuracy. The results demonstrated that the proposed model achieved high precision in detecting tumor boundaries and segmenting abnormal brain tissues. This research highlights the importance of segmentation models in visualizing tumor regions and assisting clinical analysis.

[5] R. Preetha et al. (2023) [4] conducted a comparative study on different deep neural network architectures for brain tumor segmentation using MRI images. The research analyzed the performance of multiple deep learning models in terms of segmentation accuracy, computational complexity, and efficiency.

The study emphasized that encoder–decoder architectures such as U-Net are highly effective in medical image segmentation tasks due to their ability to preserve spatial information through skip connections. The comparative analysis provided valuable insights into selecting suitable deep learning models for tumor segmentation and highlighted the importance of combining classification and segmentation techniques for improved diagnostic performance.

Research Gap

Although several deep learning techniques have been developed for brain tumor detection and segmentation, certain limitations still exist in current systems. The main research gaps identified from the literature are:

- Many existing systems focus only on tumor classification or segmentation individually, rather than integrating both tasks within a single framework.
- Several deep learning models require high computational resources, making them difficult to deploy on compact or embedded platforms.
- Most research implementations are designed for high-performance computing environments rather than portable diagnostic systems.
- Limited studies address real-time user interaction and visualization of diagnostic results.

Proposed Solution

To address the limitations identified in previous studies, this project proposes an integrated brain tumor detection system using deep learning techniques. The proposed system combines a **ResNet50-based classification model** with a **U-Net segmentation model** to analyze MRI brain images.

Initially, MRI images are preprocessed to improve image quality and ensure compatibility with the deep learning models. The classification model is used to determine whether the MRI image contains a tumor. If a tumor is detected, the segmentation model is applied to highlight the tumor region within the MRI image.

The system provides clear visualization of the tumor region along with the classification result through a user-friendly interface. By integrating classification, segmentation, and interactive visualization in a single framework, the proposed system aims to provide an efficient and practical diagnostic assistance tool for brain tumor detection.

Chapter 3

System Description

Brain tumor detection using Magnetic Resonance Imaging (MRI) is an important task in medical diagnosis. Traditionally, radiologists analyze MRI scans manually to identify abnormalities in the brain. However, this process can be time-consuming and may sometimes lead to interpretation errors when dealing with a large number of medical images. To improve the efficiency and reliability of diagnosis, the proposed system utilizes deep learning techniques to automatically detect brain tumors from MRI images.

The developed system integrates multiple stages including MRI image acquisition, image preprocessing, tumor classification, tumor segmentation, and result visualization. Deep learning models are used to analyze MRI images and extract relevant features that help in identifying abnormal regions in the brain. By combining both classification and segmentation techniques, the system is capable of determining whether a tumor is present and also highlighting the tumor region within the MRI image.

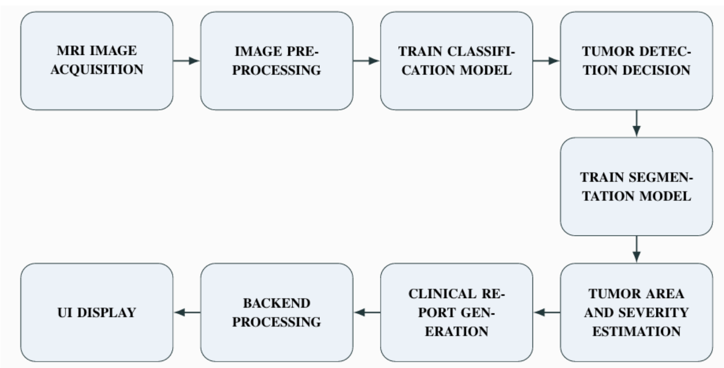
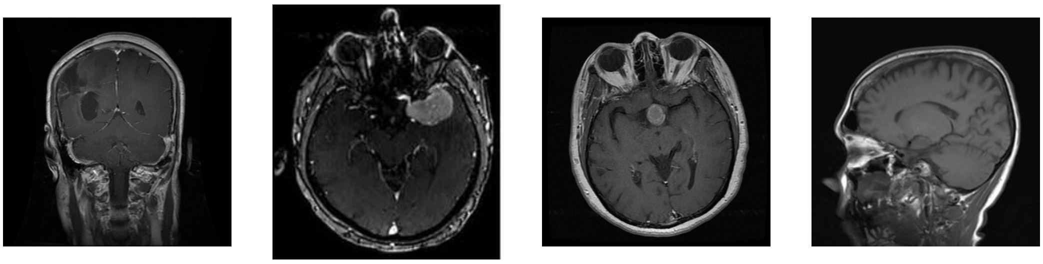


Figure 3.1: System workflow .

3.0.1 MRI Image Acquisition

The first stage of the proposed system involves acquiring Magnetic Resonance Imaging (MRI) brain images that serve as the input for tumor detection. MRI is widely used in medical diagnostics as it provides high-resolution images of soft tissues and internal brain structures without harmful radiation exposure. These images help in identifying abnormalities such as tumors, lesions, and tissue irregularities. In this project, MRI images are categorized into four classes—glioma, meningioma, pituitary tumor, and normal brain—to train the deep learning models for accurate classification. Gliomas are aggressive tumors originating from glial cells, meningiomas arise from the protective layers of the brain and are usually benign, pituitary tumors develop in the pituitary gland affecting hormone regulation, and the normal class represents healthy brain images. These categorized MRI images form the dataset used for training and testing the proposed system.



(a) Glioma (b) Meningioma (c) Pituitary (d) NoTumor

Figure 3.0.1: Sample MRI Images of Glioma, Meningioma, Pituitary and NoTumor

3.0.2 Image Preprocessing

Before feeding MRI images into the deep learning models, preprocessing is performed to improve image quality and ensure compatibility with model input requirements. One of the main steps is **image resizing**, where all MRI images are converted to a fixed size . This normal resizing ensures uniform input, reduces computational complexity, and matches the input size required by the model.

Another step is **normalization**, where pixel values are scaled (usually between 0 and 1) to improve model training stability and performance. The images are also converted into numerical arrays so that they can be processed by neural networks.

Thus, the preprocessing techniques used include resizing, normalization, and image-to-array conversion, which help in improving the accuracy and efficiency of the model.

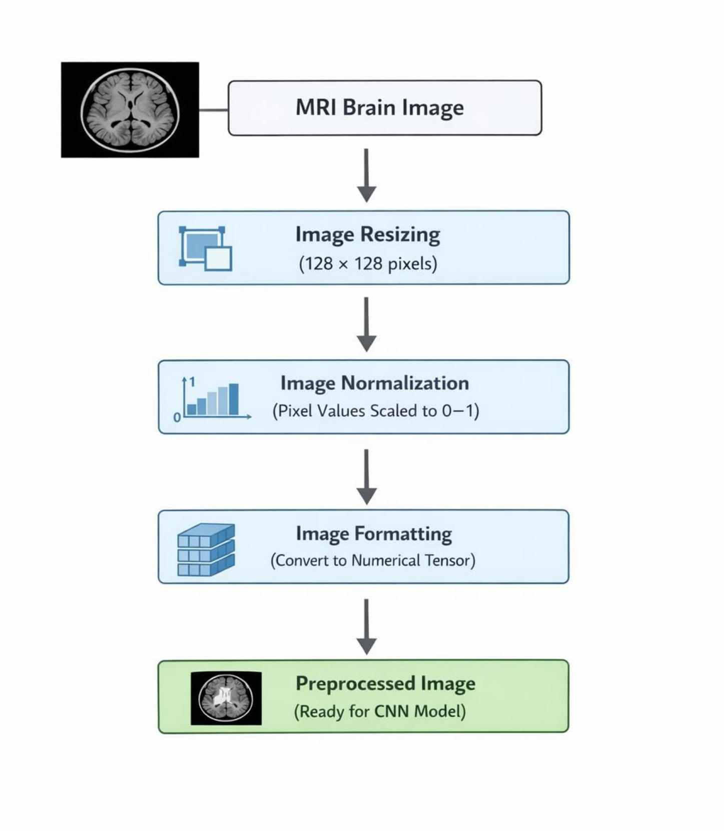


Figure 3.0.2: Preprocessing

3.0.3 Tumor Classification

After preprocessing, the MRI images are provided to a classification model to determine whether the image contains a tumor. The classification stage is responsible for distinguishing between tumor and non-tumor images by analyzing patterns and features extracted from the MRI scans.

The classification model used in the proposed system is based on the ResNet50 architecture, which is a deep convolutional neural network designed for high-performance image classification tasks. ResNet50 introduces residual learning through shortcut connections that allow the network to train deeper architectures without suffering from the vanishing gradient problem.

The model extracts spatial features from the MRI images and processes them through multiple convolutional layers. The final output layer uses the Softmax function to generate probability values for each class and determine whether the MRI image contains a tumor or not.

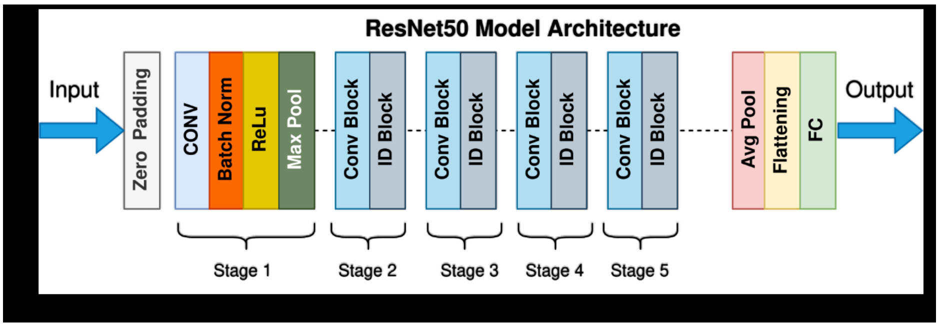


Figure 3.0.3: ResNet50 Architecture Used for Tumor Classification.

3.0.4 Tumor Segmentation

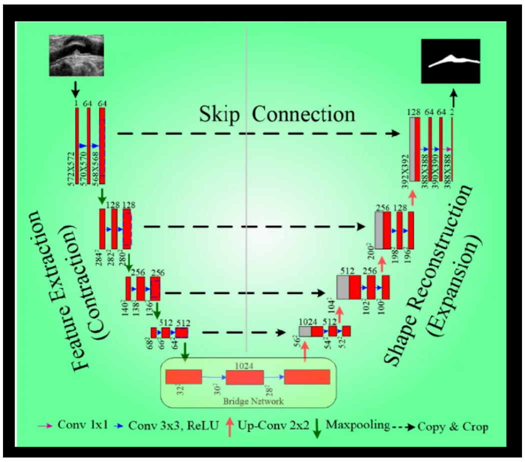


Figure 3.0.4: U-Net Architecture for Tumor Segmentation.

If the classification model detects the presence of a tumor, the system proceeds to the segmentation stage. Tumor segmentation is used to identify and highlight the exact region of the tumor within the MRI image.

The segmentation model used in the system is based on the U-Net architecture, which is widely used for biomedical image segmentation tasks. U-Net follows an encoder–decoder structure that captures both contextual and spatial information from the input image. The encoder extracts features from the image while the decoder reconstructs the segmentation mask highlighting the tumor region.

The segmentation output helps visualize the tumor region clearly and allows the system to separate abnormal tissues from the surrounding healthy brain tissues.

3.0.5 Tumor Area and Severity Estimation

After the segmentation process, the tumor region is identified as a binary mask where the tumor pixels are separated from the background. The segmented mask is used to estimate the tumor area by calculating the total number of pixels belonging to the tumor region. Since each pixel corresponds to a specific spatial resolution in the MRI image, the total tumor area can be estimated by summing the tumor pixels and multiplying by the pixel area. This quantitative measurement helps in assessing the extent of abnormal tissue present in the MRI scan.

Let $M(x, y)$ represent the segmentation mask where:

$$M(x, y) = \begin{cases} 1, & \text{if the pixel belongs to the tumor region} \\ 0, & \text{otherwise} \end{cases} \tag{3.1}$$

The tumor area A_{tumor} is calculated as:

$$A_{tumor} = \sum_{x=1}^{W} \sum_{y=1}^{H} M(x, y) \tag{3.2}$$

where W and H represent the width and height of the image respectively.

If the spatial resolution of each pixel is known, the actual tumor area can be estimated as:

$$A_{actual} = A_{tumor} \times (pixel\_size)^2 \tag{3.3}$$

To estimate the severity of the tumor, the ratio between the tumor area and the total brain region area is computed. This provides an estimate of how much of the brain region is affected by abnormal tissue.

Let A_{brain} denote the total brain area in the MRI image. The tumor severity percentage is calculated as:

$$Severity(\%) = \frac{A_{tumor}}{A_{brain}} \times 100 \tag{3.4}$$

Based on the calculated percentage, the severity level can be categorized into different ranges such as mild, moderate, or severe. This quantitative estimation assists in understanding the extent of the tumor and its impact on the surrounding brain tissue.

3.0.6 Result Visualization

The final stage of the system involves displaying the diagnostic results to the user. The system presents both the classification result and the segmented tumor region through a graphical interface.

The visualization allows users to easily interpret the results and observe the highlighted tumor region within the MRI image. By integrating classification and segmentation techniques within a single framework, the proposed system provides an effective and efficient approach for automated brain tumor detection and visualization.

Chapter 4

Hardware Description

The hardware components used in the proposed system provide the computational platform required for running the brain tumor detection application and displaying the results.The main hardware components used in the system are the Raspberry Pi 4 Model B and a 7-inch capacitive touchscreen display. These components together form a compact embedded platform capable of running the developed medical image analysis system.

4.1 Raspberry Pi 4 Model B

The **Raspberry Pi 4 Model B** serves as the primary processing unit of the proposed system.The Raspberry Pi acts as the central controller that manages the interaction between the graphical user interface and the backend processing system.

In this project, it handles MRI image input, performs model inference, and displays the analysis results through the connected touchscreen interface.

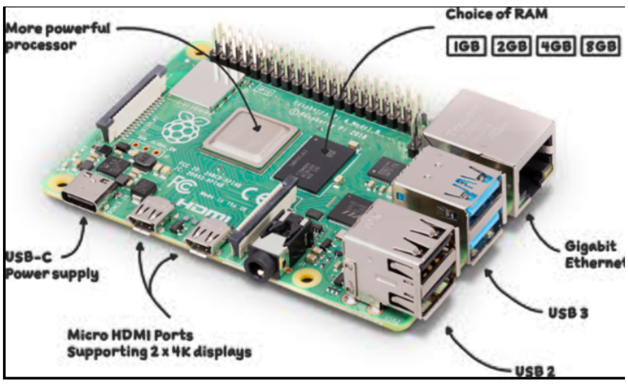


Figure 4.1: Raspberry Pi 4 Model B

Specifications

- **Processor:** Broadcom BCM2711, Quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5 GHz
- **Memory:** 2 GB, 4 GB, or 8 GB LPDDR4-3200 SDRAM
- **Storage:** MicroSD card slot used for operating system and data storage
- **Connectivity:**
  - Gigabit Ethernet
  - Dual-band 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless
  - Bluetooth 5.0 with BLE support
- **USB Ports:**
  - 2 × USB 3.0 ports
  - 2 × USB 2.0 ports
- **Video Output:**
  - 2 × Micro HDMI ports (supports up to 4K resolution at 60 fps)
  - Hardware video decoding for H.265 (4Kp60) and H.264 (1080p60 decode)
- **Audio Output:** 3.5 mm audio jack and HDMI audio output
- **GPIO Pins:** 40-pin GPIO header for connecting external peripherals
- **Power Supply:** 5 V / 3.0 A DC through USB-C connector
- **Operating System:** Raspberry Pi OS, Ubuntu, and other Linux-based distributions
- **Dimensions:** 85 mm × 56 mm × 17 mm
- **Weight:** Approximately 46 g

4.2 Touchscreen Display

The system uses a Waveshare 7-inch HDMI Capacitive Touchscreen Display with a resolution of 1024×600 pixels. This display provides an interactive interface for users to upload MRI images and view the results generated by the system.

The touchscreen display is connected to the Raspberry Pi through an HDMI interface for video output and a USB connection for touch input.

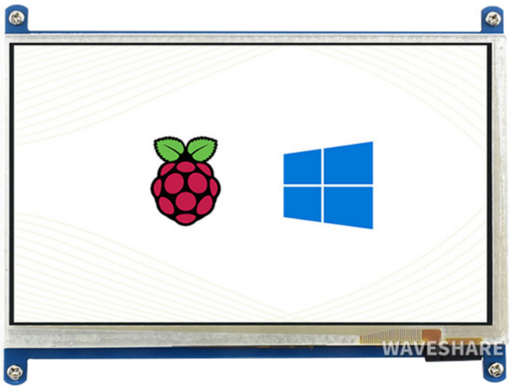


Figure 4.2: Waveshare 7-inch HDMI Capacitive Touchscreen Display

Specifications

- **Display Size:** 7-inch
- **Resolution:** 1024×600 pixels
- **Display Type:** IPS LCD panel
- **Touch Technology:** Capacitive multi-touch screen
- **Interface:** HDMI for display and USB for touch input
- **Viewing Angle:** Wide viewing angle for better visibility
- **Power Consumption:** Low power operation suitable for embedded systems
- **Compatibility:** Supports Raspberry Pi and other HDMI display devices

Chapter 5

Software Description

The software components used in the proposed system play an important role in implementing the brain tumor detection framework and enabling interaction between the user and the system. The software environment supports image processing, deep learning model execution, backend communication, and graphical user interface development. Various programming languages, frameworks, and libraries are used to build and deploy the system effectively. The software components work together to process MRI images, perform tumor detection and segmentation, and display the results through an interactive interface.

5.1 Python Programming Language

Python is the primary programming language used in the development of the proposed system. It is widely used in the fields of artificial intelligence, machine learning, and data science due to its simple syntax and extensive library support. Python provides powerful libraries for image processing, numerical computation, and deep learning model development.

In this project, Python is used to implement the deep learning models, perform MRI image preprocessing, and handle the backend processing of the application. Python also enables easy integration of different frameworks such as TensorFlow, PyTorch, and FastAPI, making it suitable for building complex AI-based systems.

5.2 TensorFlow and TensorFlow Lite

TensorFlow is an open-source deep learning framework widely used for building and training neural network models. It provides a flexible platform for designing machine learning algorithms and performing large-scale data processing. In the proposed system, TensorFlow is used to develop the classification model for detecting the presence of brain tumors in MRI images.

For deployment on embedded systems, the trained classification model is converted into TensorFlow Lite format. TensorFlow Lite is a lightweight version of TensorFlow that is optimized for edge devices such as the Raspberry Pi. It reduces the size of the model and improves inference speed while maintaining high prediction accuracy. This makes it suitable for running deep learning models efficiently on devices with limited computational resources.

5.3 PyTorch

PyTorch is another open-source deep learning framework that is widely used for research and development of neural networks. It provides a flexible and dynamic computational graph that allows developers to design and train deep learning models easily. PyTorch supports GPU acceleration and efficient tensor operations, making it suitable for complex image processing tasks.

In this project, PyTorch is used for developing and training the tumor segmentation model based on the U-Net architecture. The trained segmentation model is later converted into a format suitable for deployment on the target platform.

5.4 ONNX Runtime

ONNX (Open Neural Network Exchange) is a framework that allows deep learning models to be transferred between different platforms and frameworks. It provides a standardized format for representing machine learning models, enabling interoperability between frameworks such as TensorFlow and PyTorch.

In the proposed system, the segmentation model trained using PyTorch is exported into ONNX format. The ONNX Runtime is then used to execute the model efficiently during inference. This allows the segmentation model to run effectively on the embedded platform while maintaining good performance.

5.5 FastAPI

FastAPI is a modern web framework used for building high-performance APIs using Python. It provides a simple and efficient way to develop backend services that handle communication between the user interface and the deep learning models.

In the proposed system, FastAPI is used to create the backend server that manages the processing of MRI images. When a user uploads an MRI image through the interface, the FastAPI server receives the request, performs the necessary processing using the classification and segmentation models, and returns the results to the user interface.

5.6 HTML, CSS, and JavaScript

HTML, CSS, and JavaScript are used to develop the graphical user interface of the proposed system. These web technologies allow the creation of an interactive interface that enables users to upload MRI images and view the analysis results.

HTML is used to structure the content of the web pages, while CSS is used to design and style the interface. JavaScript is used to implement interactive elements and handle communication with the backend server. Together, these technologies provide a user-friendly interface that allows users to interact with the system easily.

5.7 Operating System

The system runs on a Linux-based operating system installed on the Raspberry Pi. Raspberry Pi OS is commonly used because it provides a stable environment for running Python applications and machine learning frameworks.

The operating system manages system resources, software libraries, and hardware communication required for executing the application. It also provides support for networking, storage management, and peripheral devices connected to the Raspberry Pi. This makes it suitable for deploying embedded AI applications such as the proposed brain tumor detection system.

Chapter 6

Results and Discussion

6.1 Overview of System Evaluation

The proposed system **NeuroScan AI – Brain Tumor Diagnostic System** was evaluated to determine its effectiveness in detecting and localizing brain tumors from MRI scans. The system integrates two deep learning components:

- ResNet50 based classification model for identifying tumor type.
- U-Net based segmentation model for detecting the tumor region.

The dataset used for evaluation is the BRISC 2025 (Brain Tumor MRI Dataset for Segmentation and Classification), which contains a total of 6,000 expert-annotated T1-weighted contrast-enhanced MRI images. The dataset is divided into 5,000 images for training and 1,000 images for testing, covering four categories:

- Glioma Tumor
- Meningioma Tumor
- Pituitary Tumor
- No Tumor

The experimental evaluation includes analysis of model training performance, classification accuracy, confusion matrix evaluation, segmentation results, system interface outputs, and prototype implementation.

6.2 Patient Intake Interface

The patient intake interface allows users to enter patient information and upload MRI scans for analysis. After uploading the MRI image, the *Begin AI Analysis* button initiates the automated diagnostic process.

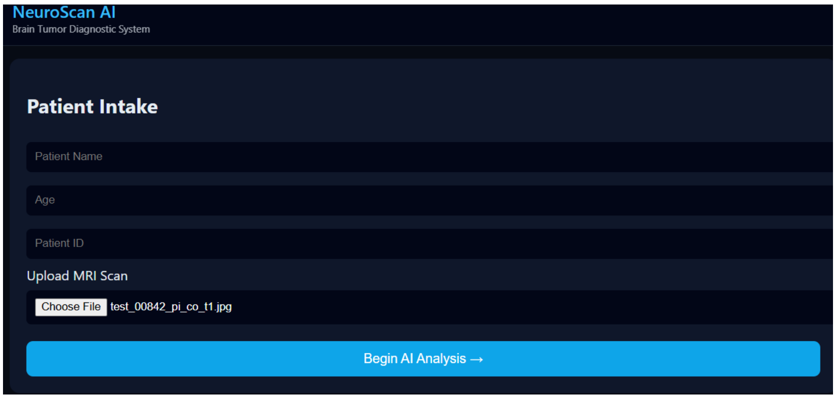


Figure 6.2: Patient Intake Interface

6.3 Classification Performance Metrics

To further evaluate the model performance, additional metrics such as precision, recall, and F1-score were calculated.

Class	Precision	Recall	F1 Score
Glioma	0.95	0.93	0.94
Meningioma	0.92	0.91	0.91
No Tumor	0.98	1.00	0.99
Pituitary	0.98	0.98	0.98

Figure 6.3: Classification Performance Metrics

The classification model achieved an overall accuracy of approximately **96%**, demonstrating reliable performance in identifying different types of brain tumors.

6.4 Confusion Matrix Analysis

To evaluate the classification performance, a confusion matrix was generated using the test dataset. The confusion matrix compares the actual class labels with the predicted labels produced by the model.

Most predictions occur along the diagonal elements of the matrix, indicating correct classification of MRI images. The No Tumor class demonstrates the highest prediction accuracy, while minor misclassification occurs between glioma and meningioma tumors due to similarities in MRI features.

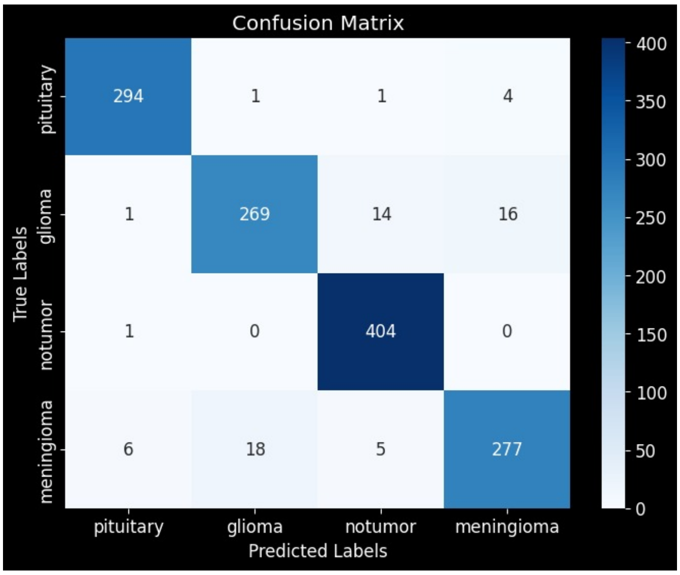


Figure 6.4: Confusion Matrix for Brain Tumor Classification

6.5 Training Performance of the Classification Model

Figure 6.5: Training Accuracy and Loss Curve

The classification model was trained using the **ResNet50 convolutional neural network architecture**. During training, the model learns important visual features from MRI images such as texture patterns and structural abnormalities associated with different tumor types.

The training process was monitored using accuracy and loss metrics across multiple epochs. The results show that the training accuracy increases progressively while the loss value decreases, indicating effective learning and convergence of the model.

6.6 Diagnosis Report

After processing the MRI image, the classification model predicts the tumor type and generates a diagnosis summary that includes tumor type, confidence score, estimated tumor area, and severity level.

Figure 6.6: Diagnosis Report Interface

6.7 Tumor Segmentation Results

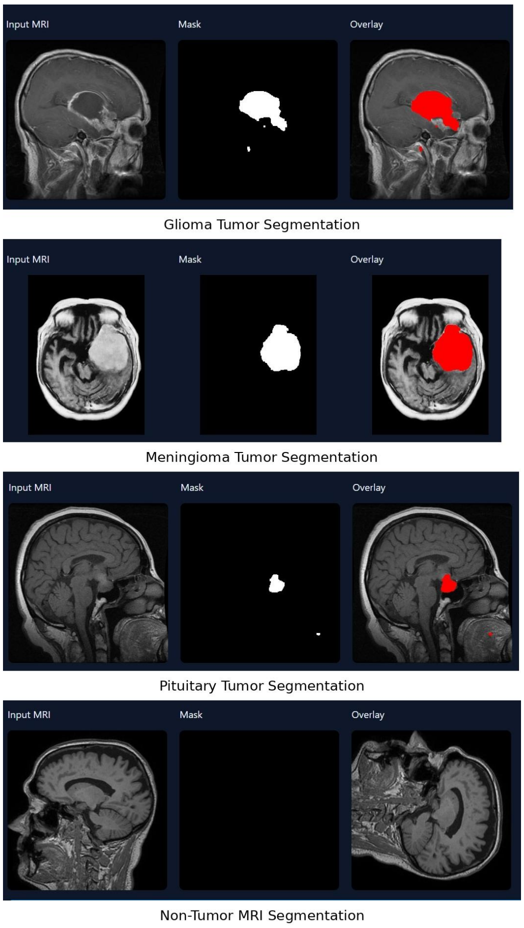


Figure 6.7: Segmentation Results for Glioma, Meningioma, Pituitary and Non-Tumor MRI Images

After classification, the segmentation module is used to identify the tumor region within the MRI image. The segmentation model generates a binary mask highlighting the tumor region.

The segmentation output consists of three components:

1. Input MRI Image
2. Predicted Binary Mask
3. Overlay Image highlighting the tumor region

In tumor MRI scans, the model successfully highlights the tumor area. For non-tumor images, the predicted mask correctly shows no highlighted regions.

6.8 Segmentation Visualization

The segmentation interface displays the tumor region detected by the model. The output includes the original MRI image, predicted mask, and an overlay highlighting the tumor location.

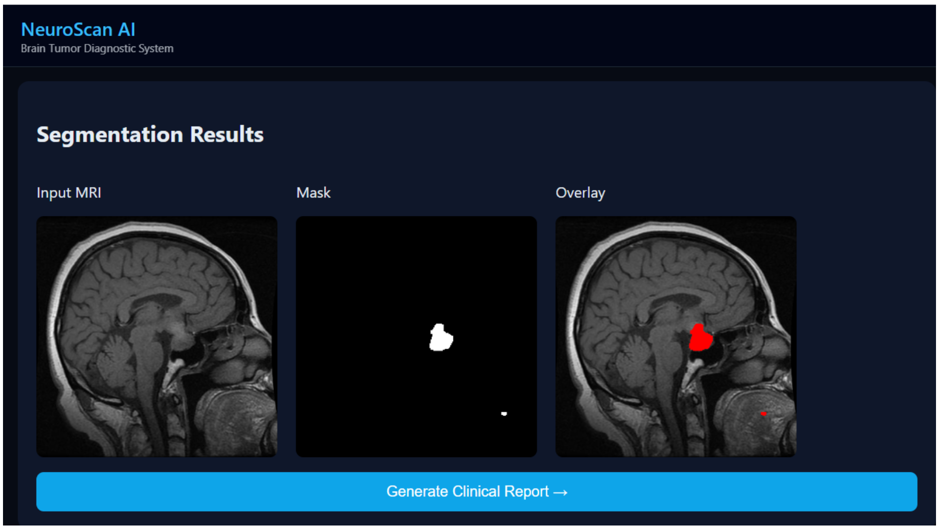


Figure 6.8: Segmentation Visualization Interface

6.9 Clinical Report Generation

The system generates an automated clinical report summarizing the AI analysis results. The report contains patient details, diagnostic findings, tumor interpretation, and recommended medical actions. The report also includes the MRI image with the AI-highlighted tumor region.

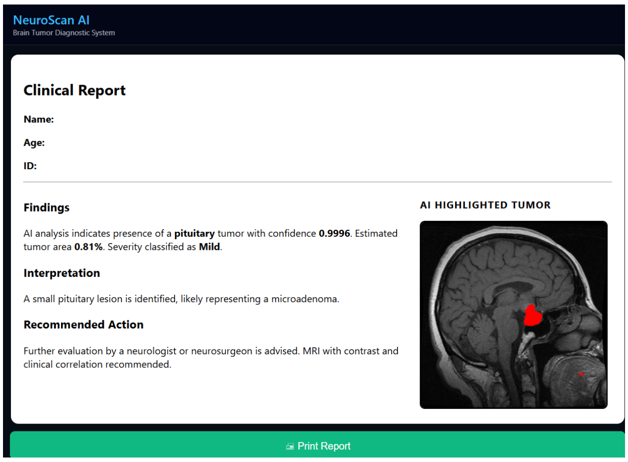


Figure 6.9: Generated Clinical Report

6.10 Prototype Implementation

The proposed system was implemented as a hardware prototype using **Raspberry Pi** integrated with the trained deep learning models and web-based interface.

The prototype demonstrates the complete workflow including MRI image upload, tumor classification, segmentation visualization, and clinical report generation. This implementation shows the feasibility of deploying the system as a compact AI-assisted diagnostic device.

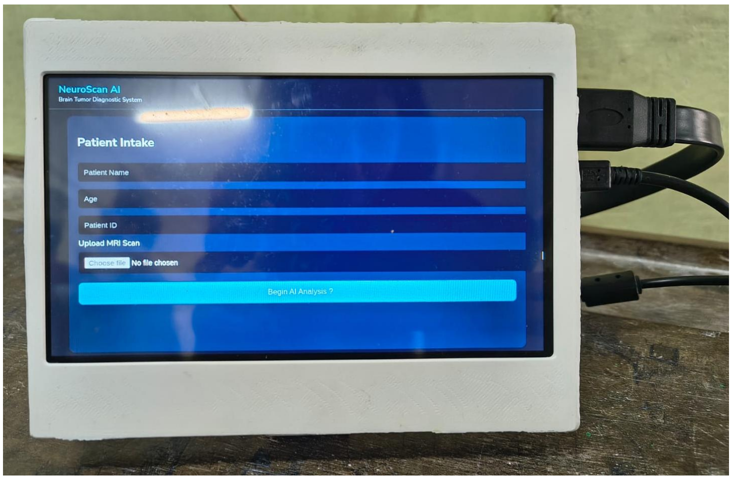


Figure 6.8: Hardware Prototype of the NeuroScan AI System

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This project presented **NeuroScan AI – Brain Tumor Diagnostic System**, an AI-based system designed to assist in the detection and localization of brain tumors using MRI images. The system integrates deep learning techniques with a user-friendly interface to provide automated diagnostic support.

A **ResNet50-based classification model** was implemented to classify MRI images into four categories: glioma tumor, meningioma tumor, pituitary tumor, and no tumor. The model achieved an overall classification accuracy of approximately **96%**, demonstrating reliable performance in identifying tumor types.

To further enhance the diagnostic capability, a **U-Net based segmentation model** was used to localize tumor regions within MRI scans. The segmentation results successfully highlighted tumor areas and generated binary masks that clearly represent the detected tumor regions.

In addition to the deep learning models, a **web-based graphical user interface** was developed to allow users to upload MRI images, view diagnostic results, and generate automated clinical reports. The system also provides visual tumor overlays and estimated tumor area to support medical interpretation.

The entire system was deployed as a **hardware prototype using Raspberry Pi**, demonstrating the feasibility of implementing an AI-assisted diagnostic tool on a compact and low-cost platform. The integration of classification, segmentation, visualization, and report generation enables a complete end-to-end workflow for brain tumor analysis.

Overall, the proposed system demonstrates the potential of artificial intelligence in improving the efficiency and accessibility of medical image analysis and supporting healthcare professionals in diagnostic decision-making.

7.2 Future Work

Although the proposed system shows promising results, several improvements can be made in future work to further enhance its performance and applicability.

Future improvements may include:

- **Larger and more diverse datasets:** Training the models on larger and multi-institutional MRI datasets could improve the robustness and generalization capability of the system.
- **3D MRI analysis:** Future systems can incorporate 3D MRI volume analysis instead of single image slices to achieve more accurate tumor detection and localization.
- **Integration and clinical validation:** The system can be integrated with hospital information systems and electronic health records, and future work may involve real-time testing and validation in clinical environments with medical professionals.

With these improvements, the proposed system could evolve into a more comprehensive AI-powered medical diagnostic platform for assisting radiologists and healthcare practitioners.

References

[1] Abdusalomov, A. B., et al. Brain tumor detection based on deep learning approaches and magnetic resonance imaging. *Cancers 15*, 16 (2023), 4172.

[2] He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2016), pp. 770–778.

[3] Neamah, K., Mohamed, F., Waheed, S. R., Kurdi, W. H. M., Taha, A. Y., and Kadhim, K. A. Utilizing deep improved resnet50 for brain tumor classification based mri. *IEEE Open Journal of the Computer Society 5* (2024), 446–456.

[4] Preetha, R., Priyadarsini, M. J. P., and Nisha, J. S. Comparative study on architecture of deep neural networks for segmentation of brain tumor using magnetic resonance images. *IEEE Access 11* (2023), 138549–138567.

[5] Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* (2015), pp. 234–241.

[6] Singh, R., Gupta, S., Bharany, S., Almogren, A., Altameem, A., and Ur Rehman, A. Ensemble deep learning models for enhanced brain tumor classification by leveraging resnet50 and efficientnet-b7 on high-resolution mri images. *IEEE Access 12* (2024), 178623–178641.

[7] Tiwary, P. K., Johri, P., Katiyar, A., and Chhipa, M. K. Deep learning-based mri brain tumor segmentation with efficientnet-enhanced unet. *IEEE Access 13* (2025), 54920–54937.